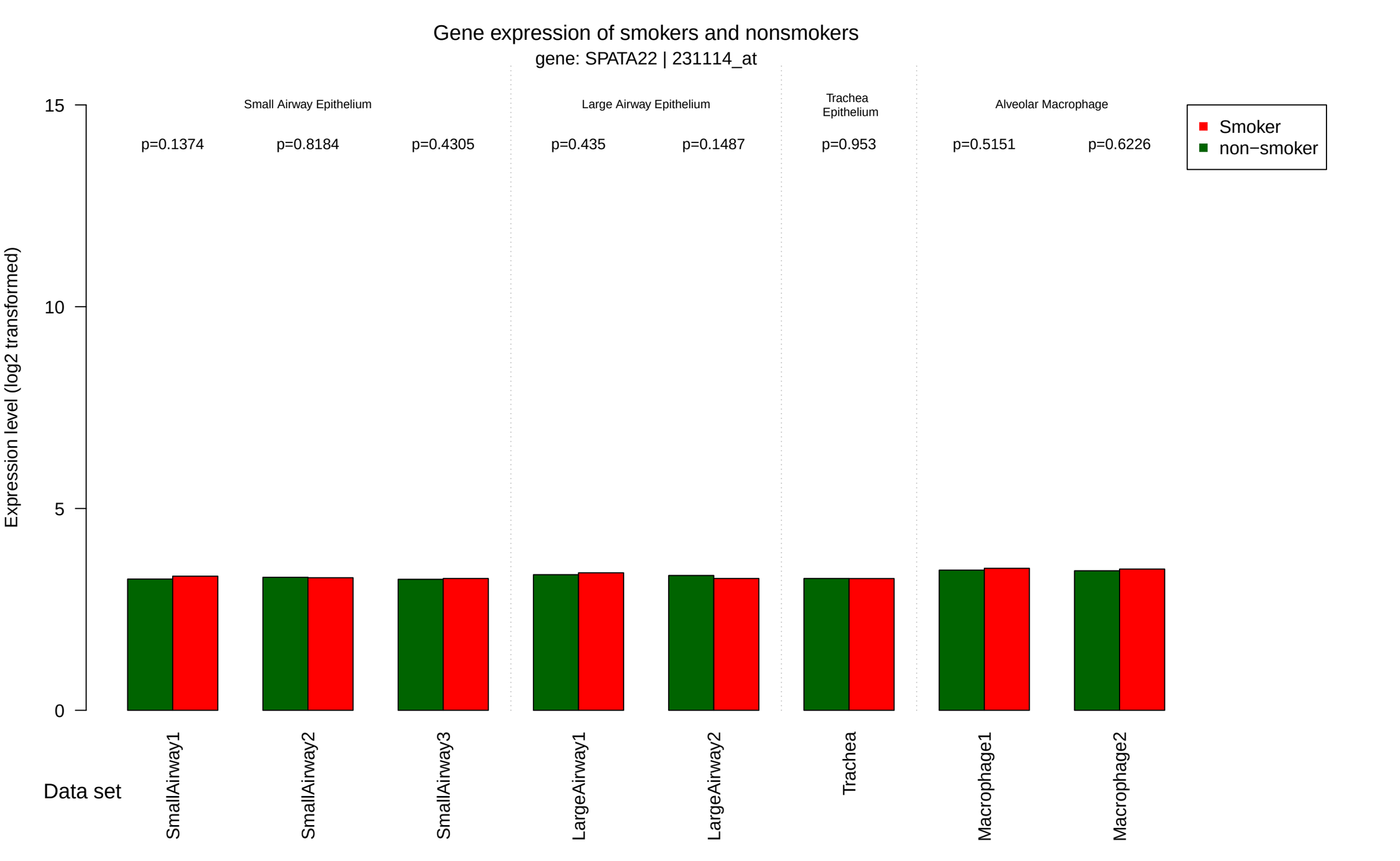
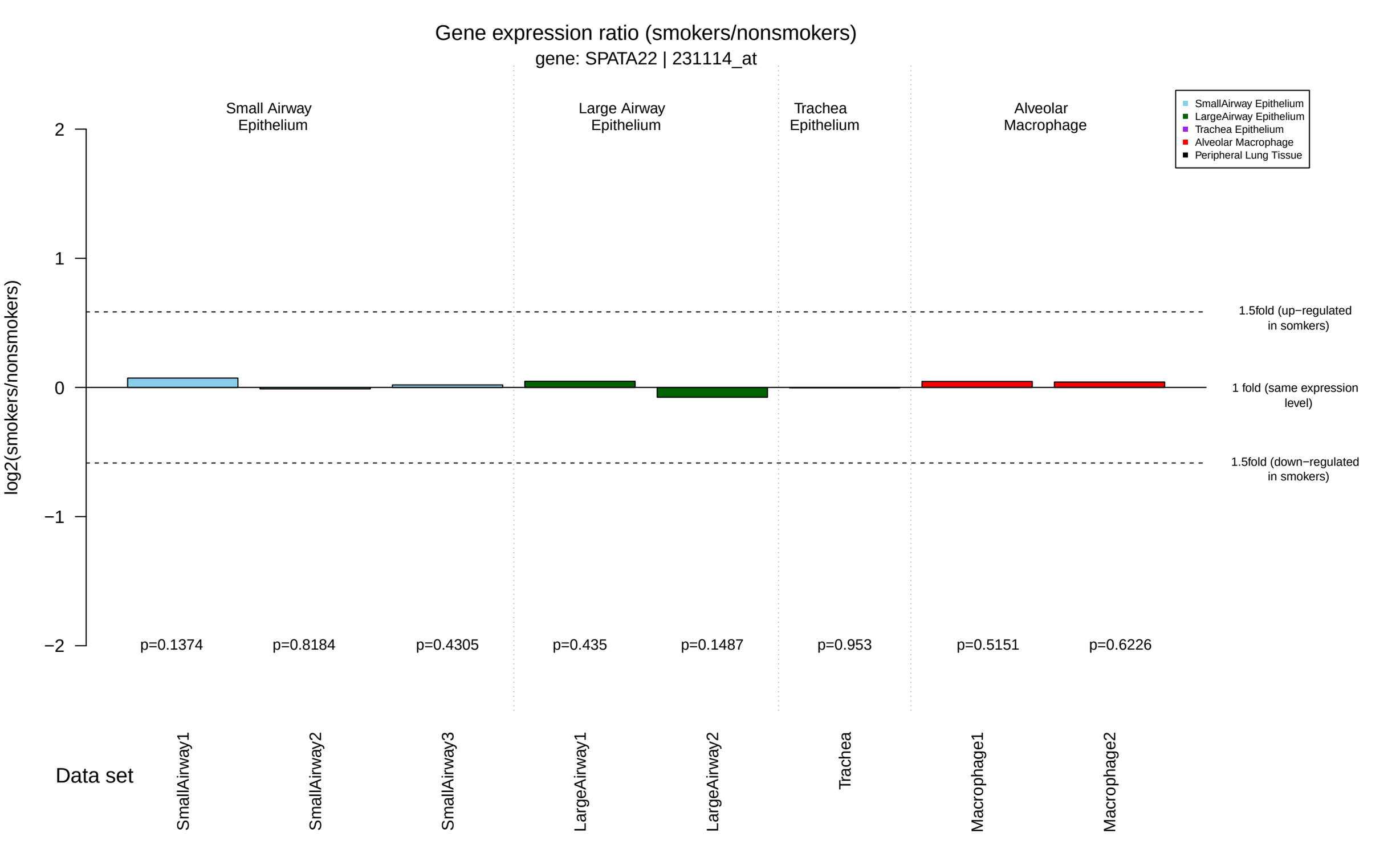
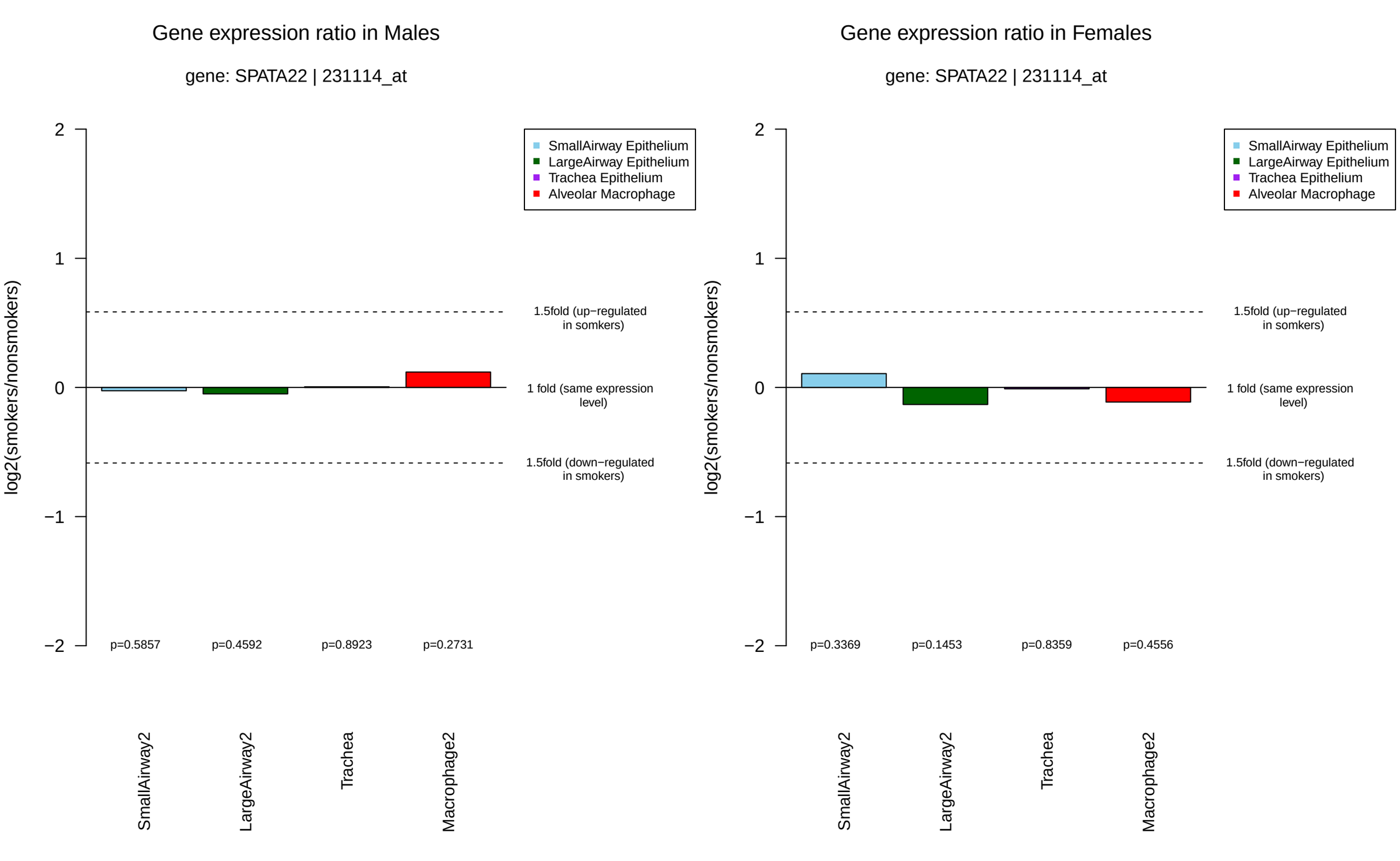
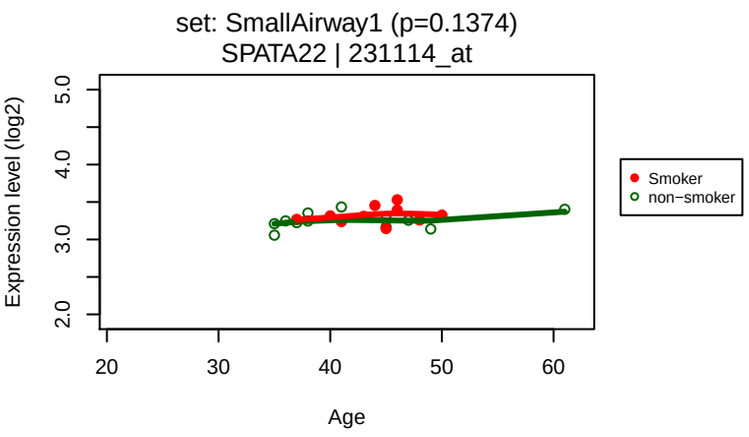
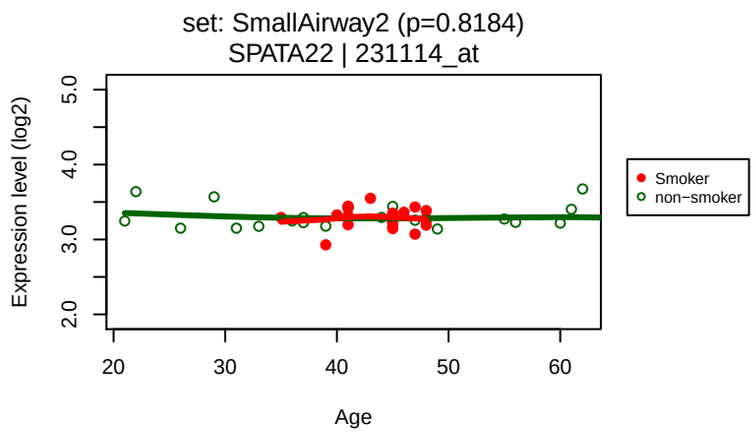
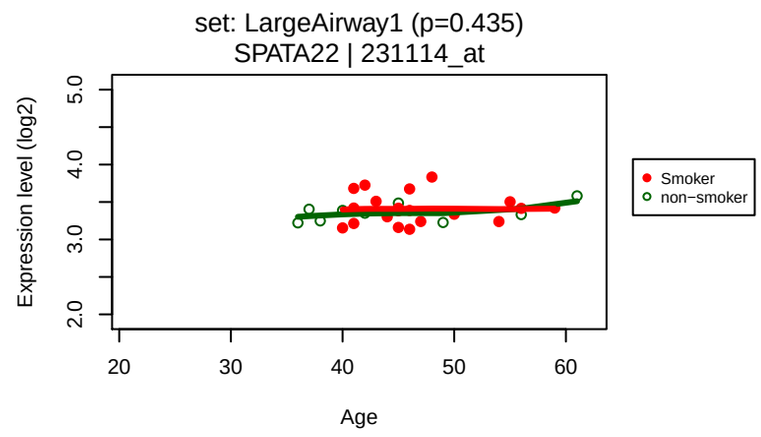

Gene expression of smokers and nonsmokers
gene: SPATA22 | 231114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1374
p=0.8184
p=0.4305
p=0.435
p=0.1487
p=0.953
p=0.5151
p=0.6226
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

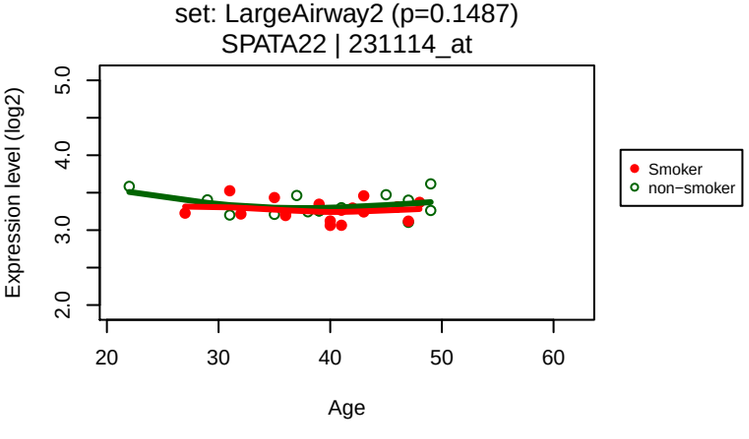

Gene expression ratio (smokers/nonsmokers)
gene: SPATA22 | 231114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1374
p=0.8184
p=0.4305
p=0.435
p=0.1487
p=0.953
p=0.5151
p=0.6226
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

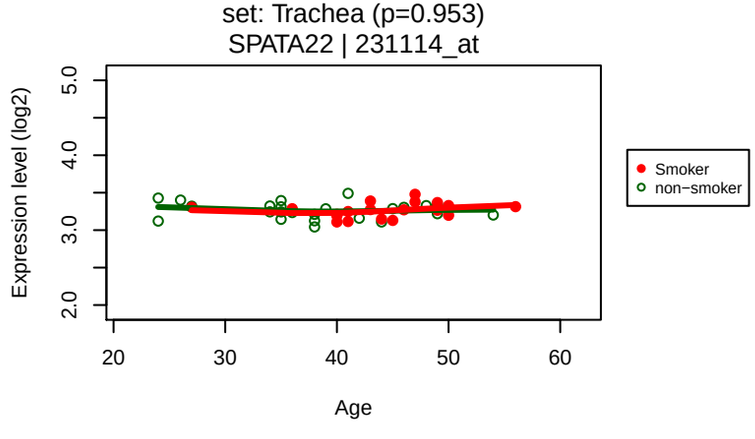

Gene expression ratio in Males
gene: SPATA22 | 231114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5857
p=0.4592
p=0.8923
p=0.2731
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA22 | 231114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3369
p=0.1453
p=0.8359
p=0.4556
SmallAirway2
LargeAirway2
Trachea
Macrophage2

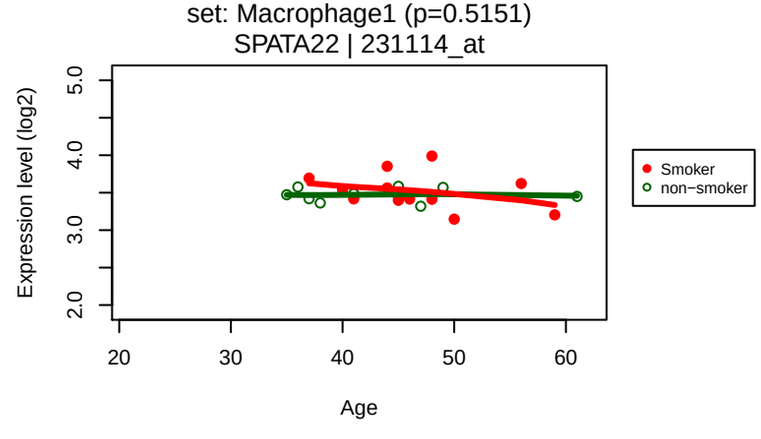

set: SmallAirway1 (p=0.1374)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker

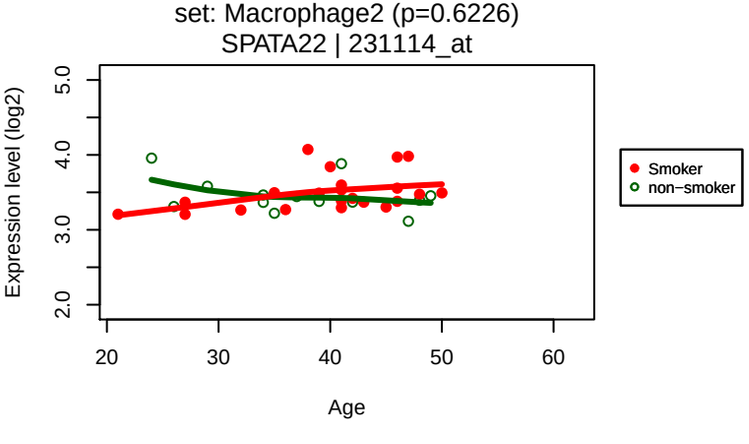

set: SmallAirway2 (p=0.8184)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker

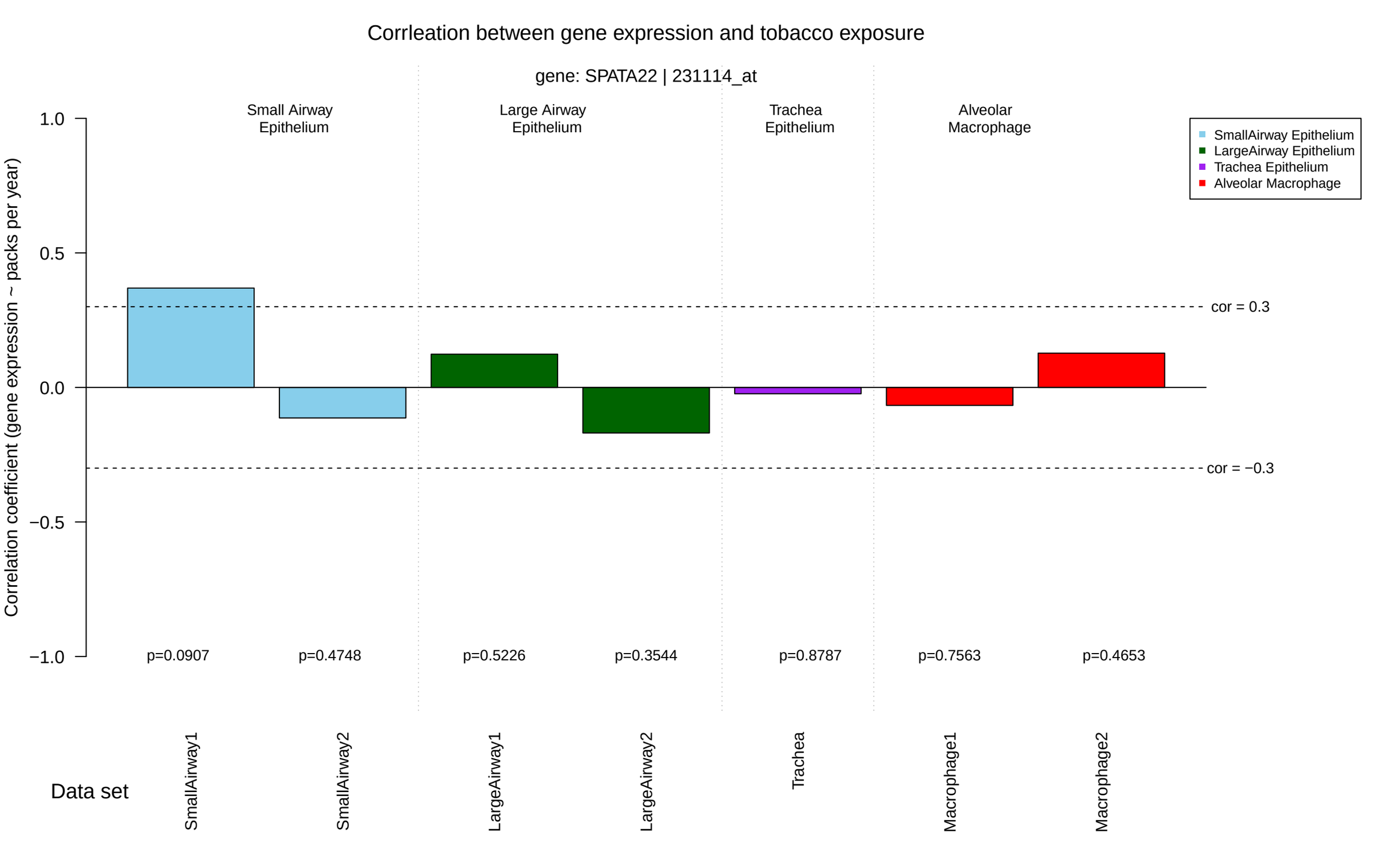

set: LargeAirway1 (p=0.435)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1487)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.953)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5151)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6226)
SPATA22 | 231114_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA22 | 231114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0907
p=0.4748
p=0.5226
p=0.3544
p=0.8787
p=0.7563
p=0.4653
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2